

# GUOCONG ZHAI

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## EDUCATION

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| <b>Ph.D. in Transportation Engineering</b><br>Old Dominion University, Norfolk, Virginia, USA<br>Dissertation: Advancing Data-Driven Decision Making for Transportation Safety: Emerging Trends, Statistical Modeling, and Causal Inference<br>Advisor: Associate Professor Kun Xie | <i>January 2021–May 2024</i>    |
| <b>M.S. in Transportation Planning and Management</b><br>Southwest Jiaotong University, Chengdu, Sichuan, China<br>Thesis: Ride-hailing User Responses to Service Suspension: A Stated Preference Analysis<br>Advisor: Professor Hongtai Yang                                       | <i>September 2017–June 2020</i> |
| <b>B.S. in Transportation Safety Engineering</b><br>Chang'an University, Xi'an, Shaanxi, China<br>Thesis: Voice Interventions and Speeding Behavior Across Driver Psychological Profiles<br>Advisor: Professor Shifeng Niu                                                          | <i>September 2012–June 2016</i> |

## APPOINTMENTS

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| <b>Assistant Professor</b><br>School of Transportation and Logistics, Southwest Jiaotong University, China                                                             | <i>May 2025–Present</i>         |
| <b>Research Fellow</b><br>Department of Civil and Environmental Engineering, National University of Singapore, Singapore<br>Mentor: Assistant Professor Prateek Bansal | <i>September 2024–July 2025</i> |
| <b>Graduate Research Assistant</b><br>Department of Civil and Environmental Engineering, Old Dominion University, USA<br>Advisor: Associate Professor Kun Xie          | <i>January 2021–August 2024</i> |
| <b>Graduate School Assistant</b><br>Graduate School, Chang'an University, China                                                                                        | <i>August 2016–June 2017</i>    |

## SELECTED RESEARCH FUNDING

|                                                                                                                                                                                                              |                                  |
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| <b>Causal Decision-Making for Safe Systems</b><br>Ministry of Education of the People's Republic of China, Overseas Postdoctoral Talent Program<br>Principal Investigator                                    | <i>2027–2029</i><br>¥930,000 CNY |
| <b>Multimodal AI-Driven Demand Identification and Route Planning for Urban Air Mobility</b><br><i>2026–2027</i><br>Fundamental Research Funds for the Central Universities<br>Principal Investigator, 2 of 2 | ¥100,000 CNY                     |
| <b>Digital Sisters: A Causally-Grounded Framework for Urban Mobility Knowledge Transfer in Data-Scarce Cities</b><br>National Natural Science Foundation of China<br>Co-Principal Investigator, 2 of 3       | <i>2026–2027</i><br>¥200,000 CNY |
| <b>Causal Inference Approach to Missing-Not-at-Random Mechanism Identification and Collaborative Imputation in Crash Data</b>                                                                                | <i>2026–2028</i>                 |

National Natural Science Foundation of China, Young Scientists Fund

Principal Investigator

¥300,000 CNY

**Home and Firm Location Choice Models and Platform Development**

2024–2027

Urban Redevelopment Authority, Singapore

Research Fellow

**Factors Influencing Pedestrian Decisions to Cross Mid-Block and Potential Countermeasures**

2022–2024

Federal Highway Administration and Virginia Department of Transportation

Research Assistant

**Improving Safety Service Patrol Performance**

2021–2023

Federal Highway Administration and Virginia Department of Transportation

Research Assistant

**Statewide Landslide Risk Assessment on Transportation Infrastructure in Virginia** 2022–2024

Old Dominion University Internal Research Grant

Research Assistant

## PUBLICATIONS

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### Theme I: Causal Inference for Transportation Safety

- [S1] Pu, Q., Xie, K., Yang, H., & Zhai, G. (2026). A vision-and-knowledge enhanced large language model for generalizable pedestrian crossing behavior inference. *under review at Travel Behaviour and Society*.
- [S2] Pu, Q., Xie, K., Zhu, Y., & Zhai, G. (2026). Generating realistic safety-critical scenarios for vehicle-pedestrian interactions. *under review at Accident Analysis and Prevention*.
- [S3] Zhai, G., Xie, K., Yang, D., & Yang, H. (2025). Do electric vehicles lead to more severe crashes? A doubly robust-based causal inference approach. *Under review at Transportation Research Part A: Policy and Practice*.
- [S4] Zhai, G., Xie, K., Yang, D., & Yang, H. (2024). Developing equity-aware safety performance functions for identifying hotspots of pedestrian-involved crashes. *Accident Analysis & Prevention*, 207, 107759.
- [S5] Zhai, G., Xie, K., Yang, H., & Yang, D. (2023). Are ride-hailing services safer than taxis? A multivariate spatial approach with accommodation of exposure uncertainty. *Accident Analysis & Prevention*, 191, 107201.
- [S6] Zhai, G., Xie, K., Yang, D., & Yang, H. (2022). Assessing the safety effectiveness of citywide speed limit reduction: A causal inference approach integrating propensity score matching and spatial difference-in-differences. *Transportation Research Part A: Policy and Practice*, 157, 94–106.
- [S7] Wang, J., Cetin, M., Yang, H., Xie, K., Zhai, G., & Ishak, S. (2024). Improving effectiveness of the safety service patrol programs: A discrete event-based simulation approach. *Transportation Research Record*, 2678(8), 1–14.
- [S8] Wu, Z., Zhai, G., & Bansal, P. (2026). Causal inference in conjoint analysis: Logit models vs. potential outcomes. *Journal of Choice Modelling* (forthcoming).
- [S9] Zhai, G., & Yang, H. (2020). Is the front passenger seat always the “death seat”? An application of a hierarchical ordered probit model for occupant injury severity. *International Journal of Injury Control and Safety Promotion*, 27(4), 438–446.

## Theme II: AI-Driven Behavioral Analytics for Sustainable Shared Mobility

- [M1] Zhai, G., Yang, H., Yang, Z., Chen, S., Jiang, L., & Khattak, A. J. (2026). Assessing the effects of congestion surcharge on taxi ridership: Disentangling spatial spillover, nonlinearity, and time trends. *Under review at Transport Policy*.
- [M2] Liu, X., Zhai, G., Yang, H., Wang, R., Wang, D., & Witlox, F. (2026). Modeling taxi-metro competition: Uncovering nonlinear influences of trip and built environment factors. *Under review at Transportation Research Part D: Transport and Environment*.
- [M3] Zhang, H., Xue, X., Zhai, G., Yang, H., Lu, X., & Cherry, C. (2026). Choice preferences for urban air mobility: A latent class discrete choice model with mixed logit extensions. *Travel Behaviour and Society*, 43, 101229.
- [M4] Yang, Z., Zhai, G., Sze, N. N., Ding, H., Bobylev, N., & Yang, H. (2026). Exploring different patterns of bike-and-ride trips and influencing factors using geographically weighted random forest. *Travel Behaviour and Society*.
- [M5] Zhai, G., Wang, R., Liu, X., Mladenović, M. N., Tang, Y., Mu, H., Liu, X., & Yang, H. (2025). Built environment impacts on zonal shared e-scooter expenses: A Bayesian learning approach. *Transportation Research Part D: Transport and Environment*, 148, 105020.
- [M6] Yang, H., Luo, P., Li, C., & Zhai, G. (2023). Nonlinear effects of fare discounts and built environment on ridesplitting adoption rates. *Transportation Research Part A: Policy and Practice*, 168, 103–118.
- [M7] Yang, H., An, P., Wei, H., & Zhai, G. (2023). Unique in the metro system: The likelihood to re-identify a metro user with limited trajectory points. *Physica A: Statistical Mechanics and Its Applications*, 610, 128–141.
- [M8] Yang, H., Zhai, G., & Yang, L. (2022). How does the suspension of ride-sourcing affect the transportation system and environment? *Transportation Research Part D: Transport and Environment*, 102, 103131.
- [M9] Zhang, Z., Zhai, G., Xie, K., & Xiao, F. (2022). Exploring the nonlinear effects of ridesharing on public transit usage: A case study of San Diego. *Journal of Transport Geography*, 104, 103449.
- [M10] Yang, H., Guo, Z., Zhai, G., Yang, L., Huo, J., & Meng, X. (2022). Exploring the spatially heterogeneous effects of the built environment on bike-sharing usage during the COVID-19 pandemic. *Journal of Advanced Transportation*, 2022, 287–301.
- [M11] Yang, H., Zhai, G., Liu, X., Yang, L., Liu, Y., & Yuan, Q. (2022). Determinants of city-level private car ownership: Effects of vehicle regulation policies and relative prices. *Transport Policy*, 115, 40–48.

## PROFESSIONAL SERVICE

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### Editorial Service

- Guest Editor, *Transportation Research Part D: Transport and Environment* (2025). Organized a special issue on *AI-Driven Behavioral Analytics for Sustainable Shared Mobility: Advancing Models, Policies, and Implementation*.
- Youth Editorial Board Member, *Digital Transportation and Safety* (2024–Present).
- Youth Editorial Board Member, *Transportation Safety and Environment* (2023–Present).

### Peer Review

Completed more than 60 peer reviews for journals including *Transportation Research Parts A–D*, *Safety Science*, *Accident Analysis & Prevention*, *Journal of Transport Geography*, *Transportation*, *Transport Policy*, and *Cities*.

Outstanding Reviewer, *Transportation Research Part D*, 2021.

## AWARDS AND HONORS

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- Academic Research Excellence Award, Old Dominion University College of Engineering (2023). Sole recipient school-wide.
- Rollie Dubbe Engineering Scholarship, Old Dominion University (2022). Sole recipient university-wide.
- Outstanding Reviewer Award, *Transportation Research Part D* (2021).
- Outstanding Graduate, Southwest Jiaotong University (2020).

## METHODS AND TECHNICAL SKILLS

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- **Causal inference & statistical learning:** propensity score methods, doubly robust estimation, Bayesian machine learning
- **Generative AI & LLM toolkits:** large language model prompting and fine-tuning, retrieval-augmented generation (RAG), vision-language models
- **Programming:** Python (Pandas, NumPy, Scikit-learn, Statsmodels, GeoPandas, PyTorch, Transformers), R, SQL, and MATLAB
- **Geographic information systems:** ArcGIS and QGIS